SAMBP Technical Report TR-74/2028

Negative-Sequence Overcurrent Protection for IBR Networks

Relay 46 Adaptation, GFM Misoperation Risk, Adaptive Polarisation: 90-Scenario Study

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1 Background and Objectives

1.1 Motivation

Negative-sequence overcurrent (Relay 46) is a standard protection function for detecting unbalanced faults (SLG, LL, DLG) in conventional power systems. The fundamental assumption is that *negative-sequence current flows from source to fault*: sources have a low negative-sequence impedance and the fault is the negative-sequence sink. This assumption breaks down in IBR networks:

- **GFL IBR:** Injects near-zero negative-sequence current (converter maintains balanced three-phase output). Relay 46 sees reduced I_2 and may fail to operate (under-detection).
- **GFM IBR:** Can inject *reverse* negative-sequence current to actively balance the network voltage (VSM virtual rotor dynamics). This can cause Relay 46 at adjacent buses to see I_2 in the wrong direction and misoperate.

1.2 Objectives

1. Model negative-sequence current injection for GFL and GFM converters.

2. Quantify Relay 46 misoperation risk for three IBR mix levels.
3. Develop an adaptive polarisation scheme using positive-sequence voltage.
4. Validate 90 scenarios across 3 IBR mix levels × 30 fault types.

2 Theory and Methodology

2.1 GFL Negative-Sequence Model

A GFL converter with balanced current control injects: $\mathbf{I}_{\text{GFL}} = [I_1, 0, 0]^T$ (sequence components), i.e., $I_2^{\text{GFL}} \approx 0$. The actual I_2 depends on control loop bandwidth and grid impedance asymmetry:

$$I_2^{\text{GFL}} = \frac{V_2}{Z_2^{\text{ctrl}} + Z_{\text{grid},2}} \tag{1}$$

where $Z_2^{\text{ctrl}} = R_{\text{ctrl}} + j\omega_0 L_{\text{ctrl}}$ is the effective negative-sequence control impedance ($\approx$ 10–50 pu for fast current controllers). The effective I_2^{GFL} is ≤ 0.03 pu for most fault conditions.

2.2 GFM Negative-Sequence Model

A GFM converter with VSM control actively suppresses negative-sequence voltage:

$$\mathbf{V}_{\text{GFM,ref}} = V_1 e^{j\theta} + V_2^* e^{-j\theta} \cdot G_{\text{NS}}(s) \tag{2}$$

where $G_{\text{NS}}(s)$ is a negative-sequence suppression filter. The resulting I_2^{GFM} flows *out* of the GFM source (toward the fault), which can reverse the apparent direction of I_2 seen at adjacent Relay 46 elements.

The reverse negative-sequence index is: $\xi_2 = \angle(I_2) - \angle(V_2) + 180°$; $\xi_2 \in (-90°, +90°)$ for forward, outside for reverse.

2.3 Adaptive Polarisation Scheme

The proposed adaptive scheme uses the positive-sequence voltage V_1 as the polarisation reference for Relay 46:

$$I_{2,\text{pol}} = I_2 \cdot e^{j(\angle V_1 + \phi_{\text{comp}})} \tag{3}$$

where $\phi_{\text{comp}} = \angle(Z_1/Z_2)$ compensates for the impedance angle difference between the positive and negative sequence networks. This polarisation is robust to GFM reverse injection because V_1 is maintained during faults by the GFM source.

Pick-up criterion: $\text{Re}[I_{2,\text{pol}}] > I_{2,\text{set}}$ AND $|I_2| > I_{2,\text{min}} = 0.05$ pu.

3 Simulation Results

3.1 90-Scenario Results

3.2 Misoperation Risk Analysis

Finding 1 (GFM injection reverses I_2 direction). *At $f_{\text{gfm}} = 1.0$, 27% of standard Relay 46 elements misoperate due to GFM reverse negative-sequence injection. The adaptive positive-sequence polarisation scheme reduces misoperation to 10%, a 63% improvement.*
Finding 2 (Residual misoperation at high GFM mix). *Even with adaptive polarisation, 10% misoperation rate at $f_{\text{gfm}} = 1.0$ indicates that Relay 46 is not fully reliable in all-GFM networks. High-impedance fault (HIF) scenarios are the primary failure mode: the $|I_2| < I_{2,\text{min}} = 0.05$ pu threshold is not exceeded for HIF at $f_{\text{gfm}} = 1.0$. A lower threshold or auxiliary HIF detector (as in TR-50) is recommended.*

Table 1: TR-74 90-scenario results by IBR mix level.

IBR mix (f_{gfm})	k_{ibr}	Scenarios	AGREE	Agree Rate	Status
Low ($f_{\text{gfm}} = 0.1$)	0.3	30	30	100%	✓
Medium ($f_{\text{gfm}} = 0.5$)	0.6	30	28	93.3%	✓
High ($f_{\text{gfm}} = 1.0$)	1.0	30	27	90.0%	✓
Total	—	**90**	**85**	**94.4%**	**PASS** ($\geq$ 94%)

Table 2: Relay 46 misoperation rate: standard vs. adaptive polarisation.

IBR mix	Standard 46 misop. rate	Adaptive 46 misop. rate	Reduction
Low ($f_{\text{gfm}} = 0.1$)	0%	0%	—
Medium ($f_{\text{gfm}} = 0.5$)	13%	6.7%	48%
High ($f_{\text{gfm}} = 1.0$)	27%	10%	63%

4 Conclusion

TR-74 demonstrates that Relay 46 requires significant adaptation for IBR-rich networks. The adaptive positive-sequence polarisation scheme reduces misoperation by 63% at high GFM penetration. The 90-scenario campaign achieves 94.4% agreement across all IBR mix levels, meeting the $\geq$ 94% criterion. Residual misoperation at all-GFM networks motivates further development in TR-77.

References

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